CHAPTER SIX / NUMBERS TO NINETY-NINE

Labels for our game boxes

We will build, read and write numbers through ninety-nine.



Adamu: Our game boxes need number labels. Everyone should have a turn!
Mr Aliyu: We need the labels forty-six, eighty-one and ninety-three.
Grace: I can make neat labels, but I must check the digits.
John: Let me work out their places before you tell me.
Aishatu: My boxes are ready. I will wait while we learn.

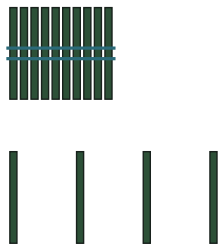
First learn in small groups of numbers. Later we will make, check and attach all three labels. Use reachable trays and large paper strips.

CHAPTER SIX / NUMBERS TO NINETY-NINE

Bring back what we know

Build first. Say the name. Write in your workbook or exercise book.

Mr Aliyu: Touch and count these strips. A tied bundle has ten separate strips. The band is not a strip.



One ten and four ones: **14, fourteen**. The left digit tells the tens. The right digit tells the loose ones.

Watch me write 14: first 1, then 4 beside it. Copy on paper. Cover it and try again.

Together, count ten loose strips, tie them, then open and count again. Make twenty with two checked bundles. Zero loose ones means 0 in the ones place.

Now try alone: count a real set of seven; build twelve and twenty. Read shuffled cards 5, 19, 2, 10, 16. Copy 12, then cover every model and your writing. An adult says a new number to write.

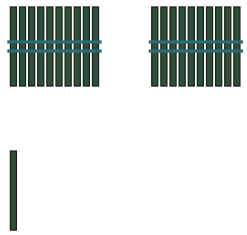
Workbook: Ready for bigger numbers. Ask for help before moving on.

CHAPTER SIX / NUMBERS TO NINETY-NINE

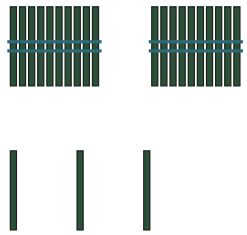
Twenty-one to twenty-five

Take several short turns. Work through each number before moving on.

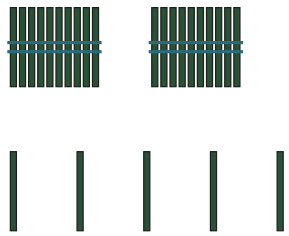
Mr Aliyu: Keep two checked tens. Add one loose strip at a time. Touch each loose one. Say the whole number. The left digit stays 2 in this group.



21 twenty-one: 2 tens and 1 one



23 twenty-three: 2 tens and 3 ones



25 twenty-five: 2 tens and 5 ones

Watch, then join in. 21 twenty-one, 22 twenty-two, 23 twenty-three, 24 twenty-four, 25 twenty-five.

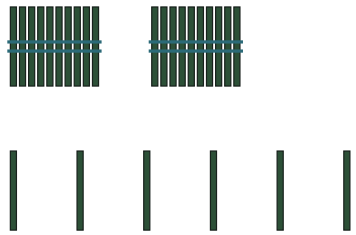
For EACH number: watch Mr Aliyu build and write it. Build and say it together; air-write, then copy on paper. Mix cards and read together. Workbook: Twenty-one to twenty-five.

CHAPTER SIX / NUMBERS TO NINETY-NINE

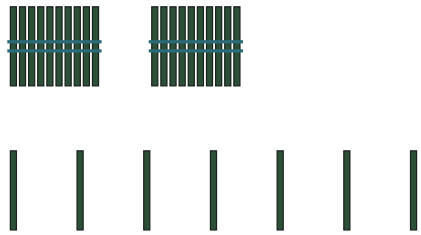
Twenty-six to twenty-nine

Take several short turns. Work through each number before moving on.

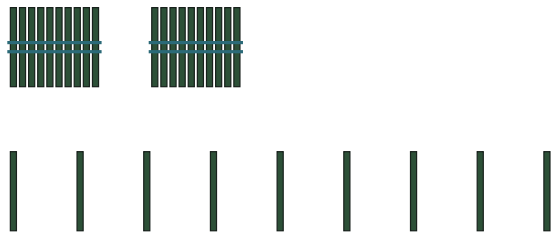
Mr Aliyu: Keep two checked tens. Add one loose strip at a time. Touch each loose one. Say the whole number. The left digit stays 2 in this group.



26 twenty-six: 2 tens and 6 ones



27 twenty-seven: 2 tens and 7 ones



29 twenty-nine: 2 tens and 9 ones

Watch, then join in. 26 twenty-six, 27 twenty-seven, 28 twenty-eight, 29 twenty-nine.

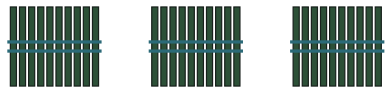
For EACH number: watch Mr Aliyu build and write it. Build and say it together; air-write, then copy on paper. Mix cards and read together. Workbook: Twenty-six to twenty-nine.

CHAPTER SIX / NUMBERS TO NINETY-NINE

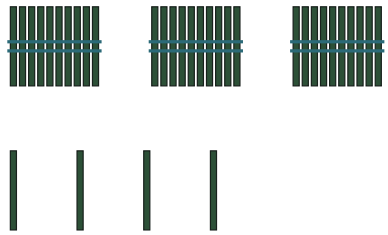
Thirty to thirty-nine

Take several short turns. Work through each number before moving on.

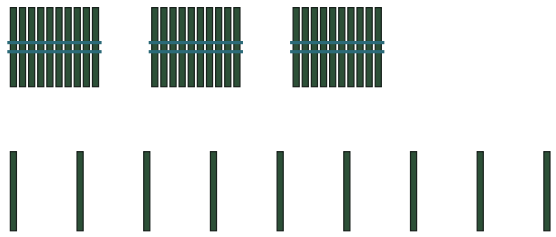
Mr Aliyu: Build 29: 2 tens and nine ones. Add one loose strip. Count the ten loose ones. Tie them as a new ten. Now there are 3 tens and no loose ones: **30, thirty**. Open the new bundle to check. Count 28, 29, 30, 31.



30 thirty: 3 tens and 0 ones



34 thirty-four: 3 tens and 4 ones



39 thirty-nine: 3 tens and 9 ones

Watch, then join in. 30 thirty, 31 thirty-one, 32 thirty-two, 33 thirty-three, 34 thirty-four, 35 thirty-five, 36 thirty-six, 37 thirty-seven, 38 thirty-eight, 39 thirty-nine.

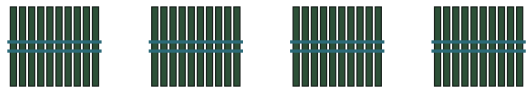
For EACH number: watch Mr Aliyu build and write it. Build and say it together; air-write, then copy on paper. Mix cards and read together. Workbook: Thirty to thirty-nine.

CHAPTER SIX / NUMBERS TO NINETY-NINE

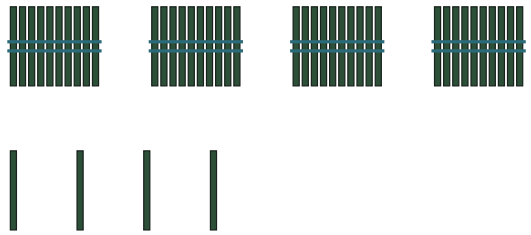
Forty to forty-nine

Take several short turns. Work through each number before moving on.

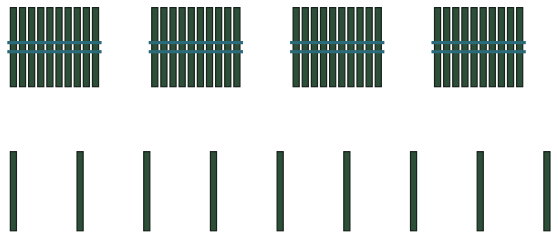
Mr Aliyu: Build 39: 3 tens and nine ones. Add one loose strip. Count the ten loose ones. Tie them as a new ten. Now there are 4 tens and no loose ones: **40, forty**. Open the new bundle to check. Count 38, 39, 40, 41.



40 forty: 4 tens and 0 ones



44 forty-four: 4 tens and 4 ones



49 forty-nine: 4 tens and 9 ones

Watch, then join in. 40 forty, 41 forty-one, 42 forty-two, 43 forty-three, 44 forty-four, 45 forty-five, 46 forty-six, 47 forty-seven, 48 forty-eight, 49 forty-nine.

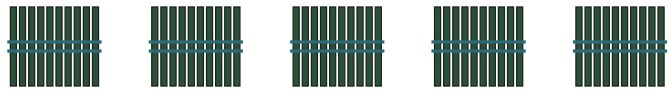
For EACH number: watch Mr Aliyu build and write it. Build and say it together; air-write, then copy on paper. Mix cards and read together. Workbook: Forty to forty-nine.

CHAPTER SIX / NUMBERS TO NINETY-NINE

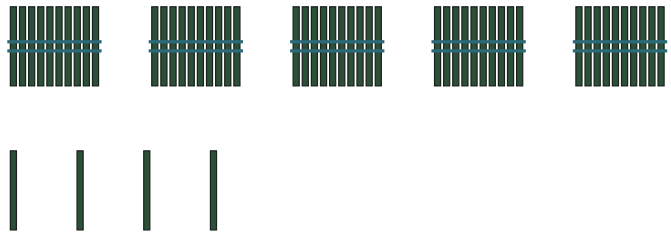
Fifty to fifty-nine

Take several short turns. Work through each number before moving on.

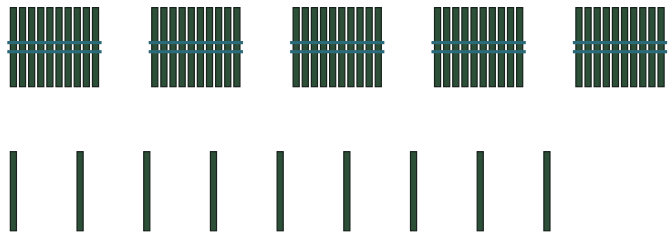
Mr Aliyu: Build 49: 4 tens and nine ones. Add one loose strip. Count the ten loose ones. Tie them as a new ten. Now there are 5 tens and no loose ones: **50, fifty**. Open the new bundle to check. Count 48, 49, 50, 51.



50 fifty: 5 tens and 0 ones



54 fifty-four: 5 tens and 4 ones



59 fifty-nine: 5 tens and 9 ones

Watch, then join in. 50 fifty, 51 fifty-one, 52 fifty-two, 53 fifty-three, 54 fifty-four, 55 fifty-five, 56 fifty-six, 57 fifty-seven, 58 fifty-eight, 59 fifty-nine.

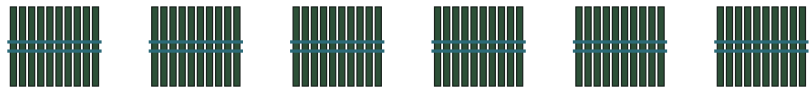
For EACH number: watch Mr Aliyu build and write it. Build and say it together; air-write, then copy on paper. Mix cards and read together. Workbook: Fifty to fifty-nine.

CHAPTER SIX / NUMBERS TO NINETY-NINE

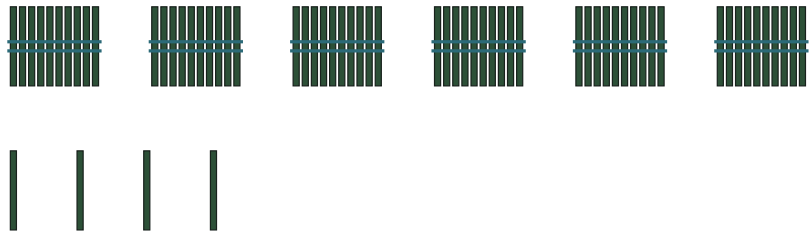
Sixty to sixty-nine

Take several short turns. Work through each number before moving on.

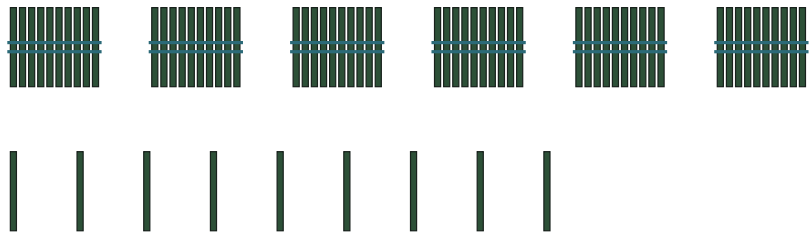
Mr Aliyu: Build 59: 5 tens and nine ones. Add one loose strip. Count the ten loose ones. Tie them as a new ten. Now there are 6 tens and no loose ones: **60, sixty**. Open the new bundle to check. Count 58, 59, 60, 61.



60 sixty: 6 tens and 0 ones



64 sixty-four: 6 tens and 4 ones



69 sixty-nine: 6 tens and 9 ones

Watch, then join in. 60 sixty, 61 sixty-one, 62 sixty-two, 63 sixty-three, 64 sixty-four, 65 sixty-five, 66 sixty-six, 67 sixty-seven, 68 sixty-eight, 69 sixty-nine.

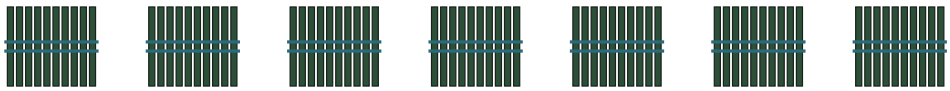
For EACH number: watch Mr Aliyu build and write it. Build and say it together; air-write, then copy on paper. Mix cards and read together. Workbook: Sixty to sixty-nine.

CHAPTER SIX / NUMBERS TO NINETY-NINE

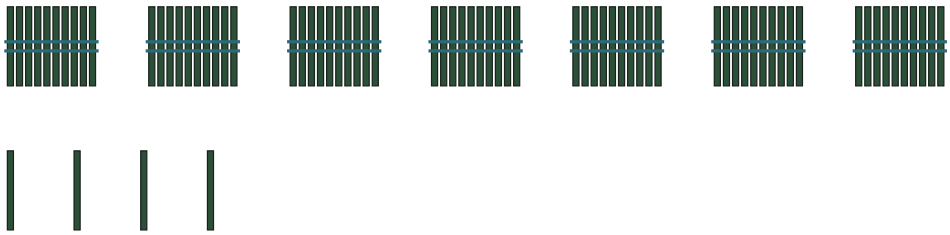
Seventy to seventy-nine

Take several short turns. Work through each number before moving on.

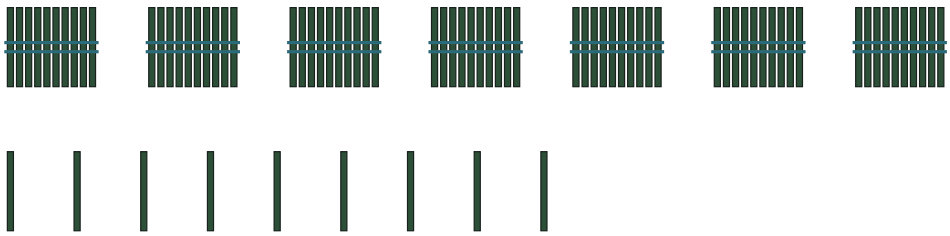
Mr Aliyu: Build 69: 6 tens and nine ones. Add one loose strip. Count the ten loose ones. Tie them as a new ten. Now there are 7 tens and no loose ones: **70, seventy**. Open the new bundle to check. Count 68, 69, 70, 71.



70 seventy: 7 tens and 0 ones



74 seventy-four: 7 tens and 4 ones



79 seventy-nine: 7 tens and 9 ones

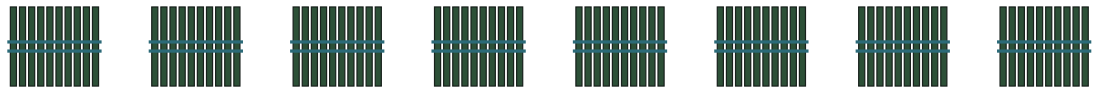
Watch, then join in. 70 seventy, 71 seventy-one, 72 seventy-two, 73 seventy-three, 74 seventy-four, 75 seventy-five, 76 seventy-six, 77 seventy-seven, 78 seventy-eight, 79 seventy-nine.
For EACH number: watch Mr Aliyu build and write it. Build and say it together; air-write, then copy on paper. Mix cards and read together. Workbook: Seventy to seventy-nine.

CHAPTER SIX / NUMBERS TO NINETY-NINE

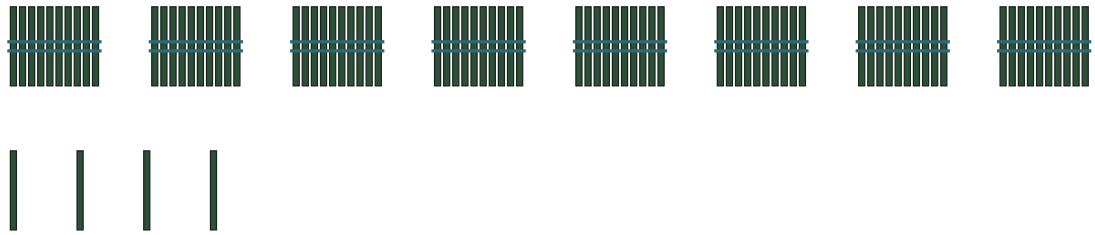
Eighty to eighty-nine

Take several short turns. Work through each number before moving on.

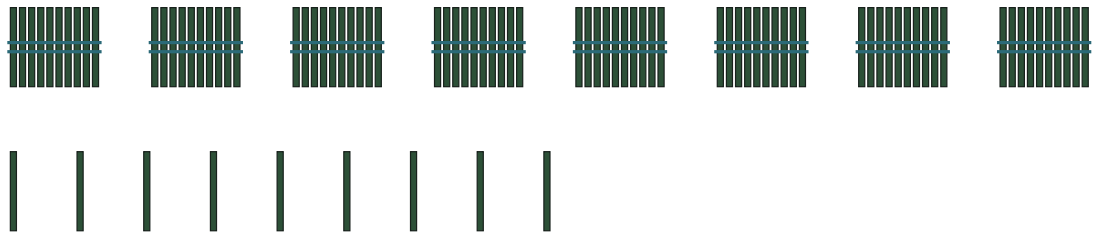
Mr Aliyu: Build 79: 7 tens and nine ones. Add one loose strip. Count the ten loose ones. Tie them as a new ten. Now there are 8 tens and no loose ones: **80, eighty**. Open the new bundle to check. Count 78, 79, 80, 81.



80 eighty: 8 tens and 0 ones



84 eighty-four: 8 tens and 4 ones



89 eighty-nine: 8 tens and 9 ones

Watch, then join in. 80 eighty, 81 eighty-one, 82 eighty-two, 83 eighty-three, 84 eighty-four, 85 eighty-five, 86 eighty-six, 87 eighty-seven, 88 eighty-eight, 89 eighty-nine.

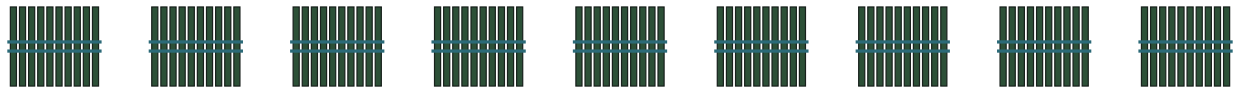
For EACH number: watch Mr Aliyu build and write it. Build and say it together; air-write, then copy on paper. Mix cards and read together. Workbook: Eighty to eighty-nine.

CHAPTER SIX / NUMBERS TO NINETY-NINE

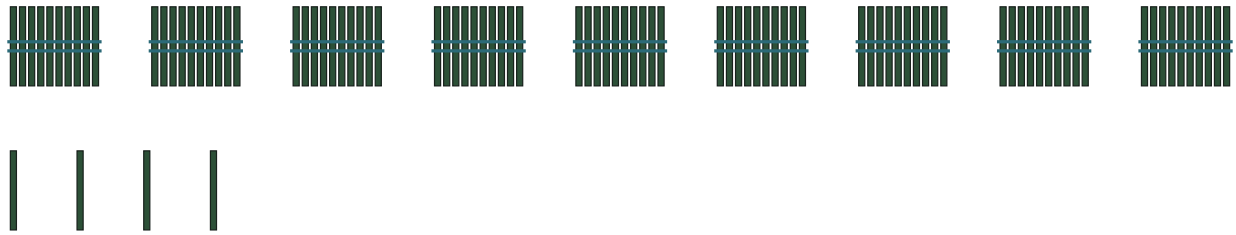
Ninety to ninety-nine

Take several short turns. Work through each number before moving on.

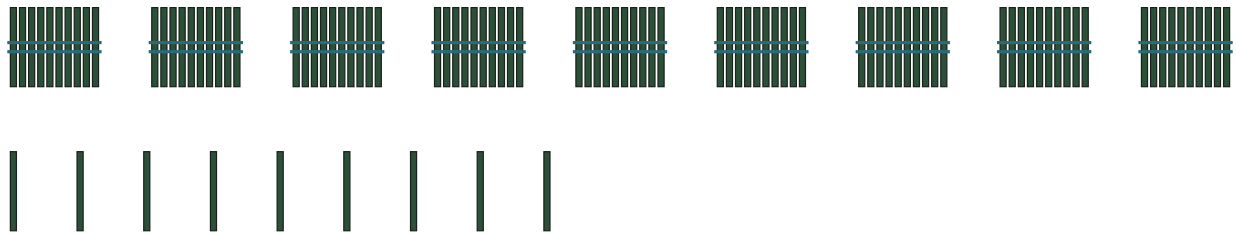
Mr Aliyu: Build 89: 8 tens and nine ones. Add one loose strip. Count the ten loose ones. Tie them as a new ten. Now there are 9 tens and no loose ones: **90, ninety**. Open the new bundle to check. Count 88, 89, 90, 91.



90 ninety: 9 tens and 0 ones



94 ninety-four: 9 tens and 4 ones



99 ninety-nine: 9 tens and 9 ones

Watch, then join in. 90 ninety, 91 ninety-one, 92 ninety-two, 93 ninety-three, 94 ninety-four, 95 ninety-five, 96 ninety-six, 97 ninety-seven, 98 ninety-eight, 99 ninety-nine.

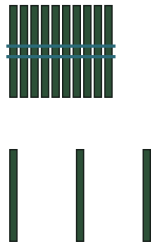
For EACH number: watch Mr Aliyu build and write it. Build and say it together; air-write, then copy on paper. Mix cards and read together. Workbook: Ninety to ninety-nine.

CHAPTER SIX / NUMBERS TO NINETY-NINE

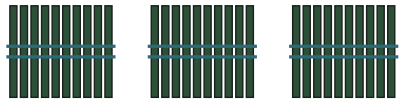
Hear the whole name

Listen, build and check. The word ending can change the number.

Mr Aliyu: Thirteen is one ten and three ones. Thirty is three tens and no loose ones. Listen: thirTEEN, THIRty. Build both, not just the first digit.



13 thirteen



30 thirty

Together build and say: fourteen / forty; fifteen / fifty; sixteen / sixty; seventeen / seventy; eighteen / eighty; nineteen / ninety. Count the tens and ones. Forty has no letter u.

Grace: I said sixty when I saw 16. I will look at both places and try again.
Adamu: I will wait, then ask you a different pair.
Adult: say one name from each pair. Children point to mixed cards, then build to check.

CHAPTER SIX / NUMBERS TO NINETY-NINE

Write places and count on

Watch Mr Aliyu. Then try together.

Mr Aliyu: In 27, 28, __, I say twenty-seven, twenty-eight, twenty-nine. Each next number has one more. I write 29 in the gap.

Together: 48, 49, __, __. Build 49. Add one, tie the ten loose strips, then add one more. Say fifty, fifty-one. Write 50, 51 on paper.

Mr Aliyu: I build 72 with seven bundles and two loose strips. I write: **7 tens and 2 ones**. Together build 84 and record its tens and ones.

Mr Aliyu: To write 31 to 33, include both ends: **31, 32, 33**. Together say and write 62 to 65. Keep the order. Check each next number has one more.

Use real strips and your exercise book, or extra workbook space. Say each whole number before you record it. Check your partner after both have tried.

CHAPTER SIX / NUMBERS TO NINETY-NINE

Try what you have learned

Try alone. Use your workbook or an ordinary exercise book.

Write forty-six in digits.

Read 81 in words.

Complete: 38, 39, __, __.

For a missing sequence: say every number in order. For example, 27, 28, __ becomes 27, 28, 29. Count on one at a time. Write your answers in the workbook or exercise book.

CHAPTER SIX / NUMBERS TO NINETY-NINE

Places, order and meaning

Try alone. Use your workbook or an ordinary exercise book.

Write 93 as tens and ones.

Write 95 to 99.

Why is 60 different from 6?

To record tens and ones, name both places: 72 is 7 tens and 2 ones. To write a range, include the first and last numbers. Example: 31 to 33 means 31, 32, 33.

CHAPTER SIX / NUMBERS TO NINETY-NINE

Remember and repair

Recall first. Get help after your own try.

How do ten loose ones become another ten? Explain or show it without looking back. Choose a number from 21 to 99. Build it. Record its tens and ones in your exercise book. Read its whole name.

Aishatu: I will let my partner finish before I check. A bundle must have ten!
John: I can explain a number in both places.

Need help? With an adult, make 19. Add one, count and tie the ten loose ones. Open the new ten and count it again. Make 29 and repeat. Then try a NEW number without help.

For a name error, build thirteen and thirty side by side. Say each name. For reversed digits, point to tens on the left and loose ones on the right. Copy once, cover it, then write a fresh spoken number.

Workbook: Remember and repair, or use an exercise book.

CHAPTER SIX / NUMBERS TO NINETY-NINE

The right labels are on

Finish the game-box job together. Then try fresh numbers alone.

Mr Aliyu: Make the three labels we needed: forty-six, eighty-one and ninety-three.
Adamu: I write 46 and give Grace a turn.
Grace: I write 81. This time I check both places.
John: I write 93 after thinking it through.
Aishatu: I check the bundles and attach the labels to my three boxes.

46 four tens, six ones
81 eight tens, one one
93 nine tens, three ones

Build each label with real strips. Check ten in every bundle. Match each written label to the spoken name. Attach 46, 81 and 93 to three different boxes. Our game boxes now have the correct labels.

A fresh try. Use new real sets, shuffled cards and spoken numbers from the adult list. Count, build, read, recognise and write separately. Keep the answer list hidden.

Workbook: A fresh number try, or fresh exercise-book space. Cover ALL earlier writing, cards and models before dictation.

FOR THE TEACHER / NUMBERS TO NINETY-NINE

Teach across short sessions

Adult guidance / Read pupil instructions aloud. Keep assessment stimuli hidden.

Prepare. Large safe strips, bands, reachable trays, three boxes, cards 1-99, paper and pencils. A bundle is ten separate strips. Open and recount it. The binding is not counted. Artwork is decorative; exact mathematics uses vector diagrams or actual strips.

Progression. Prior 1-20 check first. Re-teach any gaps before larger numbers. Use several sessions for 21-25, then 26-29, then each decade through 99. For EVERY number in a band, model build/name/formation, build/read/write together, then independent samples. Do not present the band list as one rapid lesson. All numerals use already-taught digits 0-9; demonstrate live formation and spacing again as needed.

Distributed WB checks. Use these band lists in order. Read cards: 25/21/24;29/26/28;39/30/38;49/40/48;59/50/58;69/60/68;79/70/78;89/80/88;99/90/98. Recognise the middle card among:21/22/25;26/27/29;30/31/39;40/41/49;50/51/59;60/61/69;70/71/79;80/81/89;90/91/99. Dictation lists and build/count answers follow on the next page. Cover ALL cards, models and earlier handwriting before dictation. Use fresh space or extra paper. Record count/build/read/recognition/copy/memory-writing separately. Read-aloud cards are visual stimuli, so remain visible only during reading/recognition.

Prior check. Count 7; build 12 and 20; read 5,19,2,10,16; recognise spoken 19 among 12,19,20; after copying 12 cover it and dictate fifteen (15). These are prerequisite samples, not complete 1-20 mastery. Later rotate all earlier values 1-20 across oral counting, mixed reading and covered writing.

Access. Speaking, pointing or moving strips can show understanding. An adult may scribe explanations. Record assisted handwriting separately; do not mistake a motor difficulty for a number error.

FOR THE TEACHER / NUMBERS TO NINETY-NINE

Complete task and model answers

Adult use / All six historical prompts and answer meanings are retained.

Source answers. Q1 46. Q2 eighty-one (spoken acceptable). Q3 40, 41. Q4 9 tens and 3 ones. Q5 95,96,97,98,99 in order, including both endpoints. Q6 60 is six tens (sixty ones); 6 is six ones. Practical bundles or clear explanation are valid.

Models/guided. 14=one ten/four ones; twenty=two tens/no ones. For every two-digit model, count the bundles to check the tens digit and count the loose strips to check the ones digit. Read the printed whole name. Nine loose ones plus one makes ten loose ones; exchanging makes one more ten and zero loose ones. At transitions 29/39/49/59/69/79/89 become 30/40/50/60/70/80/90. Sequence model 29; guided gaps 50,51; 72 is 7 tens/2 ones; 84 is 8 tens/4 ones; range 62,63,64,65.

Every band WB. Build/count totals respectively: 22/23, 27/28, 31/32, 41/42, 51/52, 61/62, 71/72, 81/82, 91/92. Accept clear bundles of ten and remaining ones. Read cards in each stated order with the correct whole names. Recognition targets:22,27,31,41,51,61,71,81,91. Dictation answers: 24,25; 28,29; 33,39; 43,49; 53,59; 63,69; 73,79; 83,89; 93,99. Copied numbers must match each model; memory trials must be covered.

Names/recall/story. Teen/tens pairs have one ten plus the stated loose ones versus 3-9 tens/no ones; fourteen/forty, fifteen/fifty, sixteen/sixty, seventeen/seventy, eighteen/eighty, nineteen/ninety. For any chosen number21-99, bundles and loose ones must total that number and its whole name must be correct. Recall: count/check ten, tie, retain same total. Ending labels 46,81,93 match 4t6o,8t1o,9t3o and spoken names; actually attach after checking.

FOR THE TEACHER / NUMBERS TO NINETY-NINE

Help, recheck and return later

Adult use / Keep new stimuli off the pupil page.

Fresh checks. Count a prepared 47-strip set (4 checked tens,7 loose). Build spoken thirty-six and eighty-two (3t6o;8t2o). Read shuffled 24,39,40,57,68,70,86,99 aloud: twenty-four, thirty-nine, forty, fifty-seven, sixty-eight, seventy, eighty-six, ninety-nine. Recognise spoken fifty-three among cards 35,53,50. Hide ALL cards, printed models and earlier handwriting; dictate twenty-three, thirty-eight, forty, fifty-six, sixty-seven, seventy-one, eighty-five, ninety-nine: 23,38,40,56,67,71,85,99. Use extra paper; split into brief sessions.

Misconceptions and intervention. A bundle mistaken for one: open and count ten. Reversed digits: build 24, touch two tens/four ones, record places. Teen/tens confusion: contrast 13 and 30 physically; stress whole name. After 29, child says 210: tie the ten loose ones; record 3 tens/0 ones. Zero ignored: compare 60 with 6 physically. Repeat with support, then privately recheck NEW 34 (3t4o), 50 (5t0o), read 17/70, dictate thirty-two (32) after covering. Do not award independent success for the coached example.

Delayed recall next session. Build 58 (5t8o); count prepared 64 (6t4o); read shuffled 31,49,50,62,79,80,95: thirty-one, forty-nine, fifty, sixty-two, seventy-nine, eighty, ninety-five. Recognise ninety-two among 29,92,90. Cover all models/earlier writing; dictate 29,30,44,59,60,78,90,96 using spoken names. Continue new samples in every band and 1-20 in later weeks. A small sample never establishes full-range mastery.

Record. Note independent/assisted/not yet for each mode, date and response. Copying alone does not establish memory writing; pointing alone does not establish naming. No pupil outcomes have been claimed.

FOR THE TEACHER / NUMBERS TO NINETY-NINE

Words, evidence and next steps

Adult reference / Provisional extract, not the complete annual pair.

Words. Digit: one written symbol 0-9. Numeral: written number. Ten: a group of ten ones. Ones: single items; ones place records loose ones. Tens place: left place in these two-digit numerals. Zero: no loose ones in 30,40,...,90. Sequence: numbers in an order. Bundle: tied ten, checked by opening and counting.

Curriculum map. Historical 2012 baseline P1-p03-01 identify/read numbers 1-99: prior check, each band, name contrasts, mixed fresh/delayed cards. P1-p03-02 write numerals 1-99: live formation/copying per band, covered distributed/fresh/delayed dictation; Q1/Q5. Q2 samples reading; Q3/Q4/Q6 consolidate sequence/place value. docs/PRIMARY-1-TEACHING-MAP.md and research/PRIMARY-1-ASSUMPTIONS.json provide baseline/limitations. Term 2 weeks 1-3 is editorial pacing.

Credits. Existing Mr Aliyu classroom art: artwork/redesign/scenes/mr-aliyu-blackboard-v1.png, provenance artwork/manifest.json and the matching .provenance.json sidecar. Character dress and leadership follow the series registry. All exact strips/bundles are deterministic vector marks, ten separate strips per bundle. DejaVu Sans fonts and licences in fonts/. No quantities are inferred from decorative art.

Primary 2 preview. Later work extends place value and number operations after secure Primary 1 learning. This chapter stops at 99.
Pending. Full-year assembly/revision and final four offline AI pupil pages belong to the annual book, not this extract. Current curriculum/adoption verification, local educator/pupil review, accessible-PDF certification, physical print proof and printer approval remain pending. This is not a press-ready claim.